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Rotor blade with apertured cooling air deflector

Abstract

A rotor blade includes an airfoil, a root, a cooling air passage and a cooling air deflector. The root extends axially along an axis between a first end of the root and a second end of the root. The root extends laterally between a first side of the root and a second side of the root. The root projects radially inward and away from the airfoil to an inner end of the root. The cooling air passage includes a passage inlet disposed at the inner end. The cooling air passage projects radially into the rotor blade from the passage inlet. The cooling air deflector projects radially inward from the inner end of the root to an inner end of the cooling air deflector. The cooling air deflector is disposed at the second side and is spaced laterally from the first side. The cooling air deflector includes a cooling air aperture.

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Background/Summary

TECHNICAL FIELD
(1) This disclosure relates generally to an aircraft and, more particularly, to a rotor blade for the aircraft.
BACKGROUND INFORMATION
(2) A bladed rotor of a gas turbine engine typically includes a plurality of rotor blades arranged around and mounted to a rotor disk. Various types of mounting arrangements are known in the art for mounting rotor blades to a rotor disk. Various arrangements are also known in the art for directing cooling air into one or more internal passages within a rotor blade. While these known arrangements have various benefits, there is still room in the art for improvement.
SUMMARY
(3) According to an aspect of the present disclosure, an apparatus is provided for a turbine engine. This turbine engine apparatus includes a rotor blade configured to rotate about an axis. The rotor blade includes an airfoil, a root, a cooling air passage and a cooling air deflector. The root extends axially along the axis between a first end of the root and a second end of the root. The root extends laterally between a first side of the root and a second side of the root. The root projects radially inward towards the axis and away from the airfoil to an inner end of the root. The cooling air passage includes a passage inlet disposed at the inner end of the root. The cooling air passage projects radially into the rotor blade from the passage inlet. The cooling air deflector projects radially inward towards the axis from the inner end of the root to an inner end of the cooling air deflector. The cooling air deflector is disposed at the second side of the root and is spaced laterally from the first side of the root. The cooling air deflector includes a cooling air aperture that extends laterally through the cooling air deflector.
(4) According to another aspect of the present disclosure, another apparatus is provided for a turbine engine. This turbine engine apparatus includes a rotor blade configured to rotate about an axis. The rotor blade includes an airfoil, a root, a cooling air passage and a cooling air deflector.

The root extends axially along the axis between a forward end of the root and an aft end of the root. The root extends laterally between a first side of the root and a second side of the root. The root projects radially inward towards the axis to a bottom end of the root. The cooling air passage includes a passage inlet formed in the bottom end of the root. The cooling air passage projects radially outward away from the axis into the root from the passage inlet. The cooling air deflector projects radially inward towards the axis from the bottom end of the root. The cooling air deflector is disposed laterally between the passage inlet and the second side of the root. The cooling air deflector axially overlaps the passage inlet. The cooling air deflector includes a cooling air aperture extending through the cooling air deflector from an axial end of the cooling air deflector to a lateral side of the cooling air deflector.
(5) According to still another aspect of the present disclosure, another apparatus is provided for a turbine engine. This turbine engine apparatus includes a rotor disk and a rotor blade. The rotor disk is configured to rotate about an axis and includes a slot. The rotor blade includes an airfoil, a root, a cooling air passage and a cooling air deflector. The root is received within the slow to mount the rotor blade to the rotor disk. The cooling air passage projects radially into the root from a passage inlet in a bottom end of the root. The passage inlet fluidly couples the cooling air passage to a plenum radially between the bottom end of the root and a bottom end of the slot. The cooling air deflector is configured to direct a first portion of cooling air flowing within the plenum towards the passage inlet. The cooling air deflector includes a cooling air aperture configured to direct a second portion of cooling air flowing within the plenum to a clearance gap laterally between the cooling air deflector and a sidewall of the slot.
(6) The cooling air aperture may also extend axially through the cooling air deflector.
(7) A centerline of the cooling air deflector may be angularly offset from the axis by an acute offset angle when viewed in a reference plane parallel to the inner end of the root.
(8) The acute offset angle may be between ten degrees and sixty degrees.
(9) A centerline of the cooling air deflector may be angularly offset from the axis by an acute offset angle when viewed in a reference plane perpendicular to the inner end of the root.
(10) The acute offset angle may be equal to or less than sixty degrees.
(11) A centerline of the cooling air deflector may be parallel with the axis when viewed in a reference plane perpendicular to the inner end of the root.
(12) The cooling air aperture may have a circular cross-sectional geometry when viewed in a reference plane perpendicular to a centerline of the cooling air aperture.
(13) The cooling air aperture may have a non-circular cross-sectional geometry when viewed in a reference plane perpendicular to a centerline of the cooling air aperture.
(14) The cooling air deflector may extend axially between a first end of the cooling air deflector and a second end of the cooling air deflector. The cooling air deflector may extend laterally between a first side of the cooling air deflector and a second side of the cooling air deflector. The cooling air aperture may extend laterally through the cooling air deflector from an aperture inlet to an aperture outlet. The aperture inlet may be disposed at the first end of the cooling air deflector. The aperture outlet may be disposed at the second side of the cooling air deflector.
(15) The second side of the cooling air deflector may be contiguous with the second side of the root.
(16) The first side of the cooling air deflector may be laterally next to the passage inlet.
(17) The cooling air deflector may be disposed laterally between the passage inlet and the second side of the root. The cooling air deflector may axially overlap along the passage inlet.
(18) The rotor blade may also include a flange disposed between (a) the passage inlet and the cooling air deflector and (b) the second end of the root. The flange may project radially inward towards the axis from the inner end of the root. The flange may extend laterally across the inner end of the root from the first side of the root to the second side of the root.
(19) The root may be a firtree root.

(20) The root may include a first laterally undulating surface and a second laterally undulating surface. The first laterally undulating surface may at least partially form the first side of the root. The second laterally undulating surface may at least partially form the second side of the root.
(21) The rotor blade may also include a platform radially between and connecting the airfoil and the root.
(22) The turbine engine apparatus may also include a rotor disk configured to rotate about the axis. The rotor disk may include a slot. The root may be mated with the slot to mount the rotor blade to the rotor disk. The cooling air aperture may be configured to direct cooling air to a clearance gap between the cooling air deflector and a sidewall of the slot.
(23) The present disclosure may include any one or more of the individual features disclosed above and/or below alone or in any combination thereof.
(24) The foregoing features and the operation of the invention will become more apparent in light of the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a partial side schematic illustration of a powerplant of an aircraft propulsion system.
(2) FIG. **2** is a partial end view illustration of an engine rotor.
(3) FIG. **3** is a partial sectional illustration of a rotor disk with a rotor blade in dashed line form.
(4) FIG. **4** is a partial end view illustration of a blade root of the rotor blade mated with a slot in the rotor disk.
(5) FIG. **5** is a side illustration of the rotor blade.
(6) FIG. **6** is a cross-sectional illustration of an airfoil of the rotor blade.
(7) FIG. **7** is a partial end view illustration of the rotor blade at the blade root.
(8) FIG. **8** is a perspective illustration of the rotor blade.
(9) FIG. **9** is an illustration of an inner end of the blade root.
(10) FIGS. **10**A and **10**B are illustrations of a cooling air deflector with various cooling air aperture arrangements.
(11) FIGS. **11**A and **11**B are illustrations of the cooling air aperture with various cross-sectional geometries.
DETAILED DESCRIPTION
(12) FIG. **1** illustrates a powerplant **20** of a propulsion system for an aircraft. The aircraft may be an airplane, a rotorcraft (e.g., a helicopter), a drone (e.g., an unmanned aerial vehicle (UAV)), or any other manned or unmanned aerial vehicle or system. For ease of description, the aircraft propulsion system is described below as a ducted rotor propulsion system such as a turbofan propulsion system, and the aircraft powerplant **20** is described below as a gas turbine engine **22** such as a turbofan engine. The present disclosure, however, is not limited to such an exemplary aircraft propulsion system. The aircraft propulsion system, for example, may alternatively be configured as a turbojet propulsion system, a turboprop propulsion system, a turboshaft propulsion system or an open rotor propulsion system. Moreover, the present disclosure is not limited to propulsion system applications. The turbine engine **22**, for example, may alternatively be configured as or included as part of an auxiliary power unit (APU) for the aircraft, a ground-based (e.g., industrial) electrical power system or a marine-based propulsion and/or electrical power system.
(13) The turbine engine **22** of FIG. **1** extends axially along an axis **24** between a forward, upstream end of the turbine engine **22** and an aft, downstream end of the turbine engine **22**. Briefly, the axis **24** may be a centerline axis of the turbine engine **22** and/or one or more of its members. The axis **24** may also or alternatively be a rotational axis for one or more members of the turbine engine **22**.

The turbine engine **22** of FIG. **1** includes a propulsor section **26** (e.g., a fan section), a compressor section **27**, a combustor section **28** and a turbine section **29**. The turbine section **29** includes a high pressure turbine (HPT) section **29**A and a low pressure turbine (LPT) section **29**B.

(14) The engine sections **26**-**29**B may be arranged along the axis **24** within a stationary engine housing **32**. The propulsor section **26** includes a bladed propulsor rotor **34**; e.g., a fan rotor. The compressor section **27** includes one or more bladed compressor rotors **35**. The HPT section **29**A includes a bladed high pressure turbine (HPT) rotor **36**. The LPT section **29**B includes a bladed low pressure turbine (LPT) rotor **37**; e.g., a power turbine (PT) rotor. These engine rotors **34**-**37** are housed within the engine housing **32**. The engine housing **32** of FIG. **1**, for example, includes an inner housing structure **40** (e.g., a core case structure) and an outer housing structure **42** (e.g., a propulsor case structure). The inner housing structure **40** may house one or more of the engine sections **27**-**29**B and their engine rotors **35**-**37**. The outer housing structure **42** may house at least the propulsor section **26** and its propulsor rotor **34**.

(15) The compressor rotors **35** are coupled to and rotatable with the HPT rotor **36**. The compressor rotors **35** of FIG. **1**, for example, are connected to the HPT rotor **36** through a high speed shaft **44**. At least (or only) the compressor rotors **35**, the HPT rotor **36** and the high speed shaft **44** collectively form a high speed rotating assembly **46**; e.g., a high speed spool of a core of the turbine engine **22**. This high speed rotating assembly **46** of FIG. **1** and its members **35**, **36** and **44** are rotatable about the axis **24**.

(16) The LPT rotor **37** of FIG. **1** is connected to a low speed shaft **48**. At least (or only) the LPT rotor **37** and the low speed shaft **48** collectively form a low speed rotating assembly **50**; e.g., a low speed spool of the engine core. This low speed rotating assembly **50** is further coupled to the propulsor rotor **34** through a drivetrain **52**. This drivetrain **52** may be configured as a geared drivetrain, where a geartrain **54** (e.g., a transmission, a speed change device, an epicyclic geartrain, etc.) is disposed between and operatively couples the propulsor rotor **34** to the low speed rotating assembly **50** and its LPT rotor **37**. With this arrangement, the propulsor rotor **34** may rotate at a different (e.g., slower) rotational speed than the low speed rotating assembly **50** and its LPT rotor **37**. Here, the propulsor rotor **34** and the low speed rotating assembly **50** may rotate in a common (the same) direction about the axis **24** or in opposite directions about the axis **24** depending, for example, upon the specific configuration of the geartrain **54**. Alternatively, the drivetrain **52** may be configured as a direct drive drivetrain, where the geartrain **54** is omitted. With such an arrangement, the propulsor rotor **34** rotates at a common (the same) rotational speed as the low speed rotating assembly **50** and its LPT rotor **37**. The low speed rotating assembly **50** of FIG. **1** and its members **37** and **48** as well as the propulsor rotor **34** are rotatable about the axis **24**.

(17) During operation, ambient air from outside of the aircraft enters the aircraft powerplant **20** and its turbine engine **22** through an airflow inlet **56**. This air is directed across the propulsor section **26** and into a (e.g., annular) core flowpath **58** and a (e.g., annular) bypass flowpath **60**. The core flowpath **58** of FIG. **1** extends sequentially through the compressor section **27**, the combustor section **28**, the HPT section **29**A and the LPT section **29**B from an airflow inlet **62** into the core flowpath **58** to a combustion products exhaust **64** out from the core flowpath **58** and the engine core. The air entering the core flowpath **58** may be referred to as "core air". The bypass flowpath **60** extends through a bypass duct, which bypass flowpath **60** bypasses (e.g., is disposed radially outboard of and extends along) the engine core. The air within the bypass flowpath **60** may be referred to as "bypass air".

(18) The core air is compressed by the compressor rotors **35** and is directed into a (e.g., annular) combustion chamber **66** of a (e.g., annular) combustor in the combustor section **28**. Fuel is injected into the combustion chamber **66** by one or more fuel injectors and mixed with the compressed core air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially drive rotation of the HPT rotor **36** and the LPT rotor **37** about the axis **24**. The rotation of the HPT rotor **36** drives rotation of the compressor rotors **35** about the axis

24 and, thus, compression of the air received from the core inlet **62**. The rotation of the LPT rotor **37** drives rotation of the propulsor rotor **34** through the drivetrain **52**. The rotation of the propulsor rotor **34** propels the bypass air through and out of the bypass flowpath **60**. The propulsion of the bypass air may account for a majority of thrust generated by the turbine engine **22** of FIG. **1**.

(19) FIG. **2** illustrates a portion of a bladed engine rotor **68** such as a turbine rotor. For ease of description, this engine rotor **68** is described below as being configured as or otherwise included as part of the HPT rotor **36** in FIG. **1**. It is contemplated, however, the engine rotor **68** may alternatively be configured as or otherwise included as part of the LPT rotor **37** in FIG. **1** or any other bladed rotor included in a turbine engine which utilizes cooling air. The engine rotor **68** of FIG. **2** includes a rotor disk **70** (e.g., a turbine disk) and a plurality of rotor blades **72** (e.g., turbine blades).

(20) The rotor disk **70** includes a plurality of disk slots **74**. These disk slots **74** are arranged and may be equispaced circumferentially about the axis **24** in an annular array; e.g., a circular array. Referring to FIG. **3**, each disk slot **74** extends axially along the axis **24** through the rotor disk **70** from an axial upstream end **76** of the rotor disk **70** to an axial downstream end **78** of the rotor disk **70**. However, it is contemplated this disk slot **74** may alternatively project partially axially into the rotor disk **70** from a respective one of the axial disk ends **76**, **78** in other embodiments. Here, the disk upstream end **76** is upstream of the disk downstream end **78** relative to a flow of gas (e.g., the combustion products) within the core flowpath **58** longitudinally across the rotor blades **72**. Each disk slot **74** projects partially radially into the rotor disk **70** from a radial outer periphery of the rotor disk **70** (e.g., a rim of the rotor disk **70**) to a radial inner endwall **80** (e.g., bottom end) of the respective disk slot **74**. Referring to FIG. **4**, the disk slot **74** extends laterally (e.g., circumferentially or tangentially) between and to laterally opposing sidewalls **82**A and **82**B (generally referred to as "**82**") of the respective disk slot **74**. Each of these slot sidewalls **82**A, **82**B may be at least partially or completely formed by a respective laterally undulating (e.g., a wavy, splined) surface **84**A, **84**B (generally referred to as "**84**") of the rotor disk **70**. With this arrangement, each disk slot **74** may be configured as a firtree slot. The present disclosure, however, is not limited to such an exemplary geometry. For example, in other embodiments, each disk slot **74** may alternatively be configured as a dovetail slot or a slot with another flared and/or keyed geometry.

(21) Referring to FIG. **5**, each rotor blade **72** includes an airfoil **86**, a platform **88** and a root **90**. Each rotor blade **72** of FIG. **5** also includes an internal cooling circuit **92**, a cooling air deflector **94** for the blade cooling circuit **92**, and one or more flanges **96**A and **96**B (generally referred to as "**96**").

(22) The blade airfoil **86** projects spanwise and radially outward (e.g., away from the axis **24**) out from the blade platform **88** to a tip **98** (e.g., a radial outer distal end) of the blade airfoil **86**. The blade airfoil **86** extends longitudinally along a longitudinal mean line **100** (e.g., a camber line) of the blade airfoil **86** from a leading edge **102** of the blade airfoil **86** to a trailing edge **104** of the blade airfoil **86**. Referring to FIG. **6**, the blade airfoil **86** extends widthwise between a concave, pressure side **106**A of the blade airfoil **86** and a convex, suction side **106**B of the blade airfoil **86**. These airfoil sides **106**A and **106**B (generally referred to as "**106**") extend longitudinally between and meet at the airfoil leading edge **102** and the airfoil trailing edge **104**. Referring to FIG. **5**, each of the airfoil members **102**, **104** and **106** may project spanwise and radially outward from the blade platform **88** to the airfoil tip **98**.

(23) The blade platform **88** is disposed radially between the blade airfoil **86** and the blade root **90**. The blade platform **88** of FIG. **5** is also connected to (e.g., formed integral with or otherwise attached to) the blade airfoil **86** and the blade root **90**. The blade platform **88** extends radially between a radial inner side **108** of the blade platform **88** and a radial outer side **110** of the blade platform **88**. The platform inner side **108** is disposed radially next to the blade root **90**. The platform outer side **110** is disposed radially next to the blade airfoil **86**. This platform outer side **110**

is configured to form a portion of a radial inner peripheral boundary of the core flowpath **58** longitudinally along the respective rotor blade **72**. The blade airfoil **86** thereby projects radially out from the blade platform **88** into the core flowpath **58**.

(24) The blade root **90** projects radially inwards (e.g., towards the axis **24**) out from the blade platform **88** to a distal radial inner end **112** (e.g., a bottom end) of the blade root **90**. The blade root **90** extends axially along the axis **24** from an axial upstream end **114** of the blade root **90** to an axial downstream end **116** of the blade root **90**. Here, the root upstream end **114** is upstream of the root downstream end **116** relative to the flow of gas (e.g., the combustion products) within the core flowpath **58** longitudinally across the respective rotor blade **72**. Referring to FIG. **7**, the blade root **90** extends laterally between and to opposing lateral sides **118**A and **118**B (generally referred to as "**118**") of the blade root **90**. Each of these lateral root sides **118**A, **118**B may be at least partially or completely formed by a respective laterally undulating (e.g., a wavy, splined) surface **120**A, **120**B (generally referred to as "**120**") of the blade root **90**. With this arrangement, the blade root **90** may be configured as a firtree root. The present disclosure, however, is not limited to such an exemplary geometry. For example, in other embodiments, the blade root **90** may alternatively be configured as a dovetail root or a root with another flared and/or keyed geometry.

(25) Referring to FIG. **5**, the cooling circuit **92** is configured to direct cooling air (e.g., air bled from the core flowpath **58** upstream of the combustor of FIG. **1**) into the respective rotor blade **72**, for example, to facilitate film cooling of the blade airfoil **86** and/or the blade platform **88** during turbine engine operation. The cooling circuit **92** of FIG. **5** includes one or more cooling air passages **122**A-C (generally referred to as "**122**"). Each of these cooling air passages **122**A-C projects radially into the respective rotor blade **72** and its blade root **90** from an inlet **124**A, **124**B, **124**C (generally referred to as "**124**") into the respective cooling air passage **122**A-C. More particularly, each cooling air passage **122** of FIG. **5** projects radially outward from its passage inlet **124**, through the blade root **90** and the blade platform **88**, into the blade airfoil **86**.

(26) The passage inlets **124** may be disposed at the root inner end **112**. Each passage inlet **124** of FIG. **8**, for example, is formed in a radial inner surface **126** (e.g., a planar bottom surface) of the blade root **90** at the root inner end **112**. More particularly, each passage inlet **124** of FIG. **8** pierces the root inner surface **126**. The passage inlets **124** of FIG. **8** are arranged in an axially extending array **128**. This inlet array **128** and each of its passage inlets **124** are axially spaced from the root upstream end **114** and the root downstream end **116**. The inlet array **128** and each of its passage inlets **124** are also laterally spaced from the root first side **118**A and the root second side **118**B. The inlet array **128** of FIG. **8**, for example, may be substantially axially and/or laterally centered along the root inner end **112** and its root inner surface **126**. The present disclosure, however, is not limited to such an exemplary arrangement.

(27) The cooling air deflector **94** is disposed at the root inner end **112** and connected to (e.g., formed integral with or otherwise attached to) the blade root **90**. The cooling air deflector **94** of FIG. **8**, for example, projects radially inwards out from the blade root **90** and its root inner end **112** to a distal radial inner end **130** (e.g., a bottom tip) of the cooling air deflector **94**. Referring to FIG. **9**, the cooling air deflector **94** extends axially along the axis **24** from an axial upstream end **132** of the cooling air deflector **94** to an axial downstream end **134** of the cooling air deflector **94**. The cooling air deflector **94** extends laterally between a lateral interior side **136** of the cooling air deflector **94** and a lateral exterior side **138** of the cooling air deflector **94**; see also FIG. **8**.

(28) The cooling air deflector **94** and its deflector upstream end **132** of FIG. **9** are axially spaced from the root upstream end **114** by an axial first distance. The cooling air deflector **94** and its deflector downstream end **134** of FIG. **9** are axially spaced from the root downstream end **116** by an axial second distance that is sized larger than the first distance. Here, the first distance and the second distance are sized such that the cooling air deflector **94** and its deflector interior side **136** axially overlap at least one of the passage inlets **124**; e.g., the upstream passage inlet **124**A. The cooling air deflector **94** of FIG. **9** and its deflector interior side **136**, for example, extend axially

along at least an axial major extent (e.g., at least 50%, 75% or 90%) of the upstream passage inlet **124**. The cooling air deflector **94** and its deflector interior side **136** may (or may not) also extend axially along at least an axial minor extent (e.g., less than 75%, 50% or 25%) of another one of the passage inlets **124**; e.g., the intermediate passage inlet **124**B.

(29) The cooling air deflector **94** is disposed laterally between (a) the inlet array **128** and one or more of its passage inlets **124** and (b) the root second side **118**B. The cooling air deflector **94** of FIG. **9** and its deflector interior side **136**, for example, are (e.g., slightly) axially spaced from the inlet array **128** and its passage inlets **124**A and **124**B by a lateral distance that is less than a lateral thickness of the cooling air deflector **94**. The cooling air deflector **94** or FIG. **9** and its deflector exterior side **138** are disposed at the root second side **118**B. More particularly, the deflector exterior side **138** of FIG. **9** is continuous with the root second side **118**B; see also FIGS. **7** and **8**. The cooling air deflector **94** of the present disclosure, however, is not limited to such an exemplary arrangement. The cooling air deflector **94** and its deflector interior side **136** may alternatively be arranged directly at sides of the passage inlets **124**A and **124**B in other embodiments.

(30) An upstream surface **140** of the cooling air deflector **94** at the deflector upstream end **132** is angularly offset from the axis **24** by an offset angle **142** when viewed, for example, in a first reference plane parallel with (e.g., including) the axis **24** and/or parallel with the root inner end **112** and its root inner surface **126**. The offset angle **142** is a non-zero acute angle, for example, between ten degrees (10°) and eighty degrees (80°); e.g., between thirty degrees (30°) and sixty degrees (60°). The deflector upstream end **132** and its deflector upstream surface **140** are thereby ramped laterally towards the upstream passage inlet **124**A. A downstream surface **144** of the cooling air deflector **94** at the deflector downstream end **134** may be perpendicular to the axis **24** when viewed, for example, in the first reference plane. With this arrangement, the cooling air deflector **94** axially tapers as the cooling air deflector **94** extends laterally from its deflector exterior side **138** to its deflector interior side **136**. The cooling air deflector **94** of the present disclosure, however, is not limited to such an exemplary arrangement. The deflector downstream surface **144** may alternatively be (e.g., slightly) angularly offset from the axis **24** by a non-zero acute angle in other embodiments.

(31) Referring to FIG. **8**, the deflector exterior side **138** may extent laterally along the deflector upstream end **132** and the deflector downstream end **134** and meet the deflector interior side **136** at the deflector inner end **130**. An exterior surface **146** of the cooling air deflector **94** of FIG. **8** at the deflector exterior side **138**, for example, is a curved (e.g., arcuate) surface. This deflector exterior surface **146** is contiguous with the root second surface **120**B at an interface between the cooling air deflector **94** and the blade root **90**. The deflector exterior surface **146** meets an interior surface **148** (e.g., a substantially planar surface) of the cooling air deflector **94** at its deflector interior side **136** at the deflector inner end **130**. The deflector exterior surface **146** extends axially between the deflector upstream surface **140** and the deflector downstream surface **144** both along the root second side **118**B and the deflector inner end **130**.

(32) Referring to FIG. **9**, the cooling air deflector **94** includes at least (or only) one cooling air aperture **150**. This cooling air aperture **150** extends laterally and axially, along a centerline **152** of the cooling air aperture **150**, through the cooling air deflector **94** from an inlet **154** into the cooling air aperture **150** to an outlet **156** from the cooling air aperture **150**. The aperture inlet **154** is disposed at the deflector upstream end **132** and pierces the deflector upstream surface **140**; see also FIG. **8**. The aperture outlet **156** is disposed at the deflector exterior side **138** and pierces the deflector exterior surface **146**; see also FIG. **8**. The aperture centerline **152** may follow a straight-line trajectory. The aperture centerline **152** of FIG. **9** is angularly offset from the axis **24** by an offset angle **158** when viewed, for example, in the first reference plane. The offset angle **158** is a non-zero acute angle, for example, less than seventy five degrees (75°); e.g., between ten degrees (10°) and sixty degrees (60°). Referring to FIG. **10**A, the aperture centerline **152** may be angularly offset from the axis **24** by another offset angle **160** when viewed, for example, in a second

reference plane parallel with (e.g., including) the axis **24** and/or perpendicular to the root inner end **112** and its root inner surface **126**. The offset angle **160** is a non-zero acute angle, for example, less than sixty degrees (60°); e.g., between five degrees (5°) and thirty degrees (30°). Here, the aperture centerline **152** is tilted radially outward such that the aperture outlet **156** is disposed further radially outward away from the axis **24**/radially closer to the blade root **90** than the aperture inlet **154**. Alternatively, referring to FIG. **10**B, the aperture centerline **152** may be parallel with the axis **24** when viewed, for example, in the second reference plane.

(33) Referring to FIGS. **11**A and **11**B, the cooling air aperture **150** has a cross-sectional geometry when viewed in a third reference plane perpendicular to the aperture centerline **152**. Referring to FIG. **11**A, this cross-sectional geometry may be circular. Alternatively, referring to FIG. **11**B, the cross-sectional geometry may be non-circular; e.g., oval, elliptical, oblong, etc.

(34) Referring to FIG. **9**, the flanges **96** are arranged axially along the blade root **90** between (a) the inlet array **128** and the cooling air deflector **94** and (b) the root downstream end **116**. The upstream flange **96**A of FIG. **9**, for example, is located axially between and spaced from a downstream end of the inlet array **128** and the downstream flange **96**B. The downstream flange **96**B is located at the root downstream end **116**. Referring to FIG. **8**, each of these flanges **96** is connected to (e.g., formed integral with or otherwise attached to) the blade root **90**. Each of the flanges **96** projects radially inward out from the blade root **90** and its root inner end **112** to a distal radial inner end of the respective flange **96**. Each of the flanges **96** extends laterally between and to the opposing lateral sides **118** of the blade root **90** along the root inner end **112**.

(35) Referring to FIG. **4**, the blade root **90** is mated with the disk slot **74** to mount the respective rotor blade **72** to the rotor disk **70**. Within the disk slot **74**, the root first side surface **120**A is disposed next to and substantially follows a contour of the slot first surface **84**A. Similarly, the root second side surface **120**B is disposed next to and substantially follows a contour of the slot second surface **84**B. With this arrangement, the blade root **90** radially fixes the respective rotor blade **72** to the rotor disk **70**. Here, a plenum **162** is formed by and extends radially between the blade root **90** and its root inner end **112** and the rotor disk **70** and its slot inner endwall **80**.

(36) During turbine engine operation, the cooling air is directed into the plenum **162**. This cooling air may enter the plenum **162** at the root upstream end **114**. As the engine rotor **68** rotates (e.g., counterclockwise) about the axis **24** of FIG. **4**, the cooling air may tend to be at a lateral second side of the plenum **162** when entering the plenum **162**. The cooling air deflector **94** of FIG. **9** and its deflector upstream surface **140** thereby directs a major portion (e.g., more than 90%) of the incoming cooling air laterally towards the inlet array **128** (see FIG. **8**). The cooling air may thereby be directed into the passage inlets **124** with less swirling within the plenum **162** (see FIG. **4**). By contrast, the cooling air aperture **150** of FIG. **4** may direct a minor portion (e.g., less 10%) of the incoming cooling air into a small clearance gap **164** between the cooling air deflector **94** and the slot second sidewall **82**B. By directing this minor portion of the cooling air into the clearance gap **164**, stagnant air within the clearance gap **164** may be flushed out into the plenum **162** and the deflector exterior side **138** may be cooled. In addition to the foregoing, inclusion of the cooling air aperture **150** also reduces an overall rotating weight of the respective rotor blade **72**.

(37) While various embodiments of the present disclosure have been described, it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible within the scope of the disclosure. For example, the present disclosure as described herein includes several aspects and embodiments that include particular features. Although these features may be described individually, it is within the scope of the present disclosure that some or all of these features may be combined with any one of the aspects and remain within the scope of the disclosure. Accordingly, the present disclosure is not to be restricted except in light of the attached claims and their equivalents.

Claims

1. An apparatus for a turbine engine, comprising: a rotor blade configured to rotate about an axis, the rotor blade including an airfoil, a root, a cooling air passage and a cooling air deflector; the root extending axially along the axis between a first end of the root and a second end of the root, the root extending laterally between a first side of the root and a second side of the root, and the root projecting radially inward towards the axis and away from the airfoil to an inner end of the root; the cooling air passage comprising a passage inlet disposed at the inner end of the root, and the cooling air passage projecting radially into the rotor blade from the passage inlet; the cooling air deflector projecting radially inward towards the axis from the inner end of the root to an inner end of the cooling air deflector, the cooling air deflector disposed at the second side of the root and spaced laterally from the first side of the root, and the cooling air deflector comprising a cooling air aperture that extends laterally through the cooling air deflector; the cooling air deflector extending axially between a first end of the cooling air deflector and a second end of the cooling air deflector; the cooling air deflector extending laterally between a first side of the cooling air deflector and a second side of the cooling air deflector; and the cooling air aperture extending laterally through the cooling air deflector from an aperture inlet to an aperture outlet, the aperture inlet disposed at the first end of the cooling air deflector, and the aperture outlet disposed at the second side of the cooling air deflector.
2. The apparatus of claim 1, wherein the cooling air aperture further extends axially through the cooling air deflector.
3. The apparatus of claim 1, wherein a centerline of the cooling air deflector is angularly offset from the axis by an acute offset angle when viewed in a reference plane parallel to the inner end of the root.
4. The apparatus of claim 3, wherein the acute offset angle is between ten degrees and sixty degrees.
5. The apparatus of claim 1, wherein a centerline of the cooling air deflector is angularly offset from the axis by an acute offset angle when viewed in a reference plane perpendicular to the inner end of the root.
6. The apparatus of claim 5, wherein the acute offset angle is equal to or less than sixty degrees.
7. The apparatus of claim 1, wherein a centerline of the cooling air deflector is parallel with the axis when viewed in a reference plane perpendicular to the inner end of the root.
8. The apparatus of claim 1, wherein the cooling air aperture has a circular cross-sectional geometry when viewed in a reference plane perpendicular to a centerline of the cooling air aperture.
9. The apparatus of claim 1, wherein the cooling air aperture has a non-circular cross-sectional geometry when viewed in a reference plane perpendicular to a centerline of the cooling air aperture.
10. The apparatus of claim 1, wherein the second side of the cooling air deflector is contiguous with the second side of the root.
11. The apparatus of claim 1, wherein the first side of the cooling air deflector is laterally next to the passage inlet.
12. The apparatus of claim 1, wherein the cooling air deflector is disposed laterally between the passage inlet and the second side of the root; and the cooling air deflector axially overlaps along the passage inlet.
13. The apparatus of claim 1, wherein the root is a firtree root.
14. The apparatus of claim 1, wherein the root includes a first laterally undulating surface and a second laterally undulating surface, the first laterally undulating surface at least partially forms the first side of the root, and the second laterally undulating surface at least partially forms the second

side of the root.

15. The apparatus of claim 1, wherein the rotor blade further includes a platform radially between and connecting the airfoil and the root.

16. The apparatus of claim 1, further comprising: a rotor disk configured to rotate about the axis; the rotor disk comprising a slot; the root mated with the slot to mount the rotor blade to the rotor disk; and the cooling air aperture configured to direct cooling air to a clearance gap between the cooling air deflector and a sidewall of the slot.

17. An apparatus for a turbine engine, comprising: a rotor blade configured to rotate about an axis, the rotor blade including an airfoil, a root, a cooling air passage and a cooling air deflector; the root extending axially along the axis between a first end of the root and a second end of the root, the root extending laterally between a first side of the root and a second side of the root, and the root projecting radially inward towards the axis and away from the airfoil to an inner end of the root; the cooling air passage comprising a passage inlet disposed at the inner end of the root, and the cooling air passage projecting radially into the rotor blade from the passage inlet; and the cooling air deflector projecting radially inward towards the axis from the inner end of the root to an inner end of the cooling air deflector, the cooling air deflector disposed at the second side of the root and spaced laterally from the first side of the root, and the cooling air deflector comprising a cooling air aperture that extends laterally through the cooling air deflector; wherein the rotor blade further includes a flange disposed between (a) the passage inlet and the cooling air deflector and (b) the second end of the root; wherein the flange projects radially inward towards the axis from the inner end of the root; and wherein the flange extends laterally across the inner end of the root from the first side of the root to the second side of the root.

18. An apparatus for a turbine engine, comprising: a rotor disk configured to rotate about an axis and comprising a slot; and a rotor blade including an airfoil, a root, a cooling air passage and a cooling air deflector; the root received within the slot to mount the rotor blade to the rotor disk; the cooling air passage projecting radially into the root from a passage inlet in a bottom end of the root, the passage inlet fluidly coupling the cooling air passage to a plenum radially between the bottom end of the root and a bottom end of the slot; the cooling air deflector configured to direct a first portion of cooling air flowing within the plenum towards the passage inlet; and the cooling air deflector comprising a cooling air aperture configured to direct a second portion of cooling air flowing within the plenum to a clearance gap laterally between the cooling air deflector and a sidewall of the slot.